amazon rme

AMAZON BASE BUILDING MAINTENANCE STANDARD

GLOBAL CENTRAL RME STANDARDS & COMPLIANCE

Task Title	HVAC SYSTEMS, AIR CONDENSING - 9
APM Object	HVAC.CU
APM EC	AMZ.HVAC.CU.GRP
BBM Handbook Title	Air Condition (AC unit- indoor)
General tasks always required	Follow safety procedures - e.g. 1MRA, LOTO, PPE, etc. Ensure that all WOs are carried out by qualified, certified staff with appropriate familiarisation and supervision as required Add notes for each observation and create a FWO if necessary to eliminate defects Use appropriate tools, chemicals and industry standard practices (RME policy permitting) Ensure relevant logs, readings and certification is uploaded to APM prior to WO closure These tasks are based on industry guidance, SME knowledge and HIE learning. Please ensure vendor specific requirements are followed. To suggest improvements please raise a SIM SIM - Link to SIM template

Last Update: (Version)	Jul-23	**Geo Applied:**	Intl APM Instance	**F-Gas**	Yes	**Activies may be covered by No-Go restrictions:**	Yes	**Special Certificaion Required:**	Refrigeration certificate could be required	**Delivered By:**	Carried out by trained on site technician	**OCB:**	BBM
Evidence for CT:	YES	**Pricing template Position:**	9	**Related CT Areas:**	CT.HVAC.12KW CT.HVAC.FGAS			**Assets included in this PM:**	CONDENSER UNIT	**Schedule With:**	HVAC.AC		

Description of System / Technology (According to BBM Handbook)

Air Condensing- (AC unit – outside)

An air conditioning condenser unit is the outdoor part of the AC system. The outdoor unit contains the fan, the condenser and a compressor. Both the indoor unit and the outdoor unit are required to work together to provide cool air to the rooms of your home. The condenser is the most vital part of the outdoor unit. Available options: single & multi split, VRV, VRF, etc.

Additional Information

All sites with the above listed Base Building Asset should conduct routine maintenance and inspections according to this standard PM package. The Amazon maintenance standard supports compliance, reduces safety risks,improves asset reliability & efficiency.
Some maintenances captures several assets within one PM package -> Ensure that corrective/reactive maintenance for assets where proactive maintenance is contained under another asset is created against the original asset.
This PM covers the direct evaporating outdoor unit of an AC system including heat pumps (e.g. single split, multi split, VRF, VRV, etc.)

Note : Forecast the PM for the indoor (AC.ACU) & outdoor (AC.CU) units together and combine it with regulatory tasks (e.g. leakage test F Gas regulation)

Task Index	Component	Tasks	Tasks Nested for frequencies > weekly									
			Daily	Weekly	Monthly	3 Monthly	6 Monthly	12 Monthly	24 Monthly	36 Monthly	48 Monthly	60 Monthly
2	General check & Documentation	Is the documentation of the system complete (circuit diagrams, blueprints, etc.)						PM				
3	General check & Documentation	Is the unit in operation & no alarms indicated - unit controller or control cabinet					SC	PM				
4	General check & Documentation	Are there any visual damage, missing- or loose components or corrosion					SC	PM				
5	Refrigerant leakage	Is an appropriate record for the refill of refrigerant physically available						PM				
6	Refrigerant leakage	Add to notes the refill of refrigerant during the last year in KG (year in maintenance interval, not calendar year)						PM				
7	Refrigerant leakage	Add to notes details of reason for refrigerant refill						PM				
8	AC unit check	Are there unusual noises of running compressors or condenser fans					SC	PM				
9	AC unit check	Is the condenser coil free from dirt, damage & corrosion					SC	PM				
10	AC unit check	Is there any deposit in the drip tray						PM				
11	AC unit check	Is there oil deposit visible at the connections of the pipe system					SC	PM				
12	AC unit check	Is the pipe insulation free from damage					SC	PM				
13	Functional cleaning	Clean condenser coil with state-of-the-art technology						PM				
14	Functional cleaning	Clean condenser fan blades & frame						PM				
15	Measurement	Connect the refrigerant manometer & temperature sensors (always compar with controller data)						PM				
16	Measurement	Are the refrigerant pressures and refrigerant temperatures correct						PM				
17	Measurement	Is the expansion valve working properly & superheat OK						PM				
18	Measurement	Are the condenser & evaporator temperatures ok						PM				
19	Measurement	Is the undercooling temperature ok						PM				
20	Measurement	Is the oil temperature/presuure before & after refrigerant filter ok (if appicable)						PM				
21	Measurement	Is the compressor running according to the controls (frequencies)						PM				
22	Measurement	Are the condenser fans running properly (cond. pressure ctrl., frequency, etc.)						PM				
23	Measurement	Check the compressor operation time and quantity of switching cycles if possible and react properly in case of abnormalities						PM				
24	Measurement	Add to notes operation hours of the compressors						PM				
25	Measurement	Conduct oil analyses & plan action if required (if possible)						PM				
26	Measurement	Measure the oil pressure & adjust if required (if possible)						PM				
27	Measurement	Measure the power consumption of the compressors while operating on 100% & compare the power consumption with the manufacturer specification - react in case of deviation						PM				
28	Measurement	Measure the power consumption of the condenser fans while operating on 100% & compare the power consumption with the manufacturer specification - react in case of deviation						PM				
29	Compressor & Safety	Does the low-pressure switch work properly						PM				
30	Compressor & Safety	Does the high-pressure switch work properly						PM				
31	Compressor & Safety	Simulate other safety switches of the system						PM				
32	Compressor & Safety	Is the crankcase heater working properly (attached to compressor)						PM				
33	Compressor & Safety	Is the compressor oil level ok (if visible)						PM				
34	Compressor & Safety	Is the oil separator working properly						PM				
35	Compressor & Safety	Check if there is humidity in the refrigeration system (show glass)						PM				
36	Compressor & Safety	Check refrigerant flow through show glass						PM				

37	Compressor & Safety	Check correct funcitonlity of coil defrost						PM				
38	Compressor & Safety	Does the condensate tub heater work properly						PM				
39	Final Check	Check the O&M manual and carry out additional required tasks						PM				

Change Log

July 2023 - First issue in new format. Simplification of task scope, aligmment with CT.